Squeezed superradiant lasing from a quantum many-body emitter

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Acknowledgements

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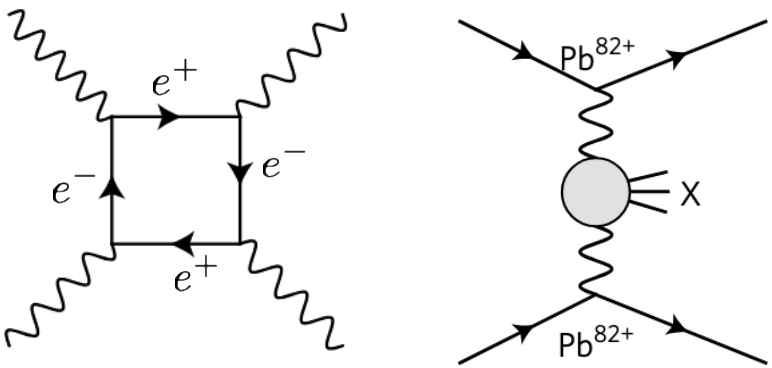
Optical coherence

Photons have **no** mutual interaction
(Landau-Yang theorem)



LHC Collaboration, Nat. Phys. 13, 852–858 (2017)

- Hard to generate coherence/correlation between photons
- Once generated, the coherence is long-lived
- Lasing is the most common method to generate coherence

A lasing device — laser

Laser——**L**ight **A**mplification by **S**timulated **E**mission of **R**adiation
Theory: Townes & Schawlow (1957); Experiment: Maiman (1960)

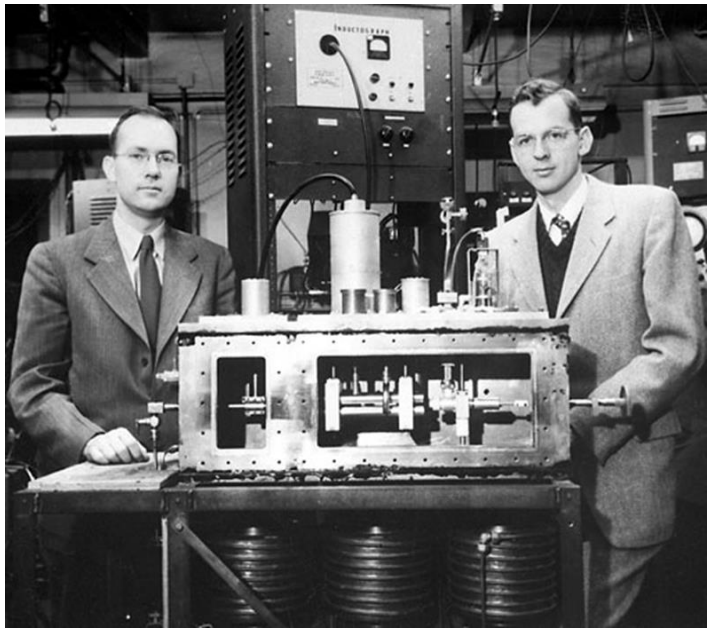
Applications:

Metrology (Spectroscopy, Interferometer,…)

Communication (Fiber, Satellite,…)

Computing (Optical Networks, Fourier Optics, …)

…

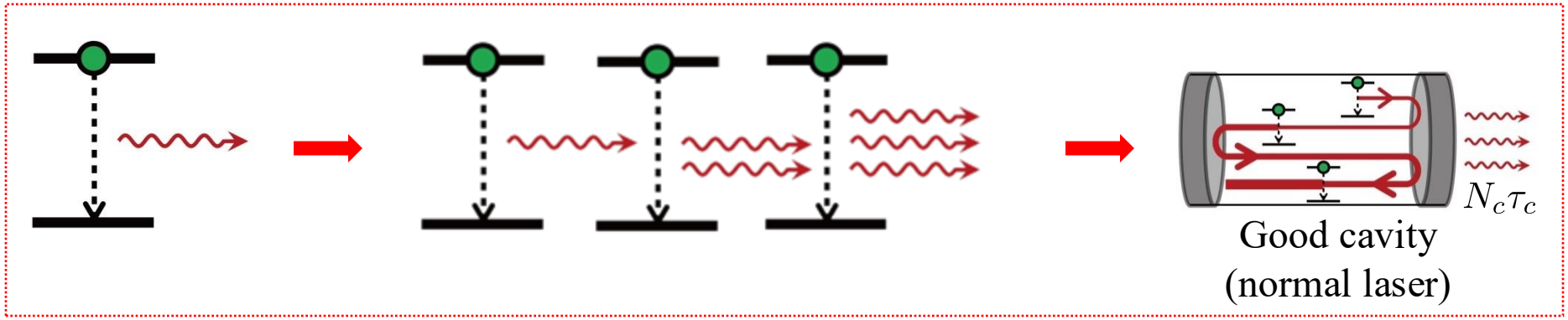


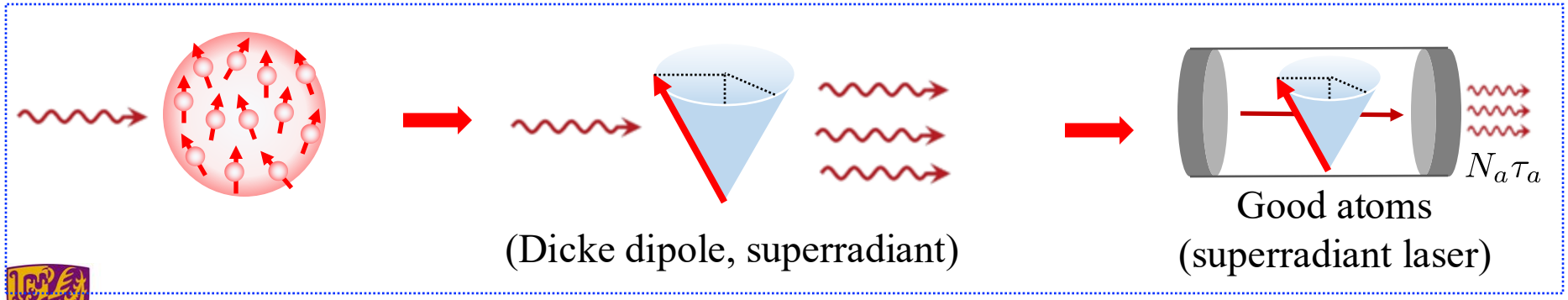
Townes (left, 1964 Nobel prize)
Inventing **maser** – prototype of **laser**

Achieve lasing

Lasing condition:

- Amplification by stimulated emission:
  population inversion: pump rate > relaxation rate
- Coherence between photons (either from cavity or from atoms)





Beyond optical coherence?

Q: In the quantum era, can we have a device generating light with "more quantum" ? (e.g., squeezed states, Fock states, cat states, entanglement, …)

Applications:

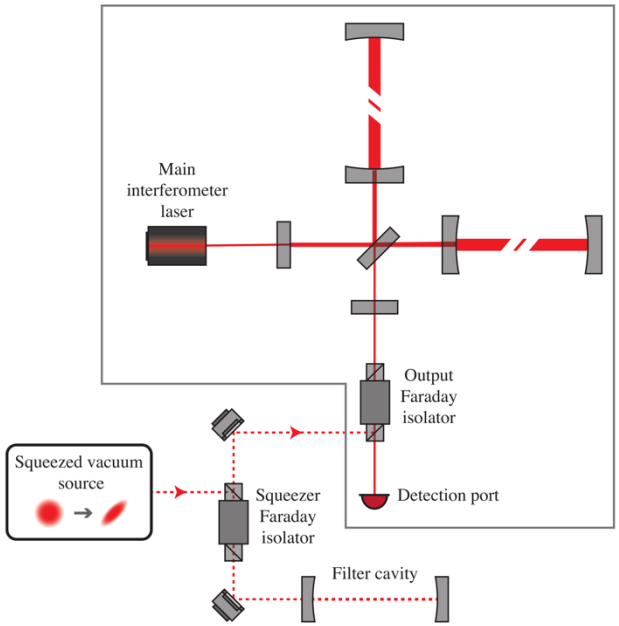
Quantum Metrology (SU(1,1), LIGO, …)

Quantum Communication (Cryptography, QKD, …)

Quantum Computing (Qubits, Gates, …)

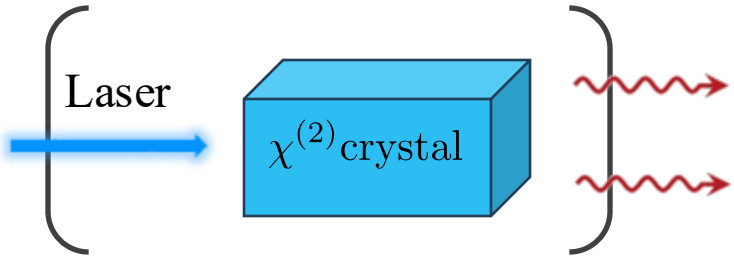
Extreme Nonlinear Optics (HHG, …)

…

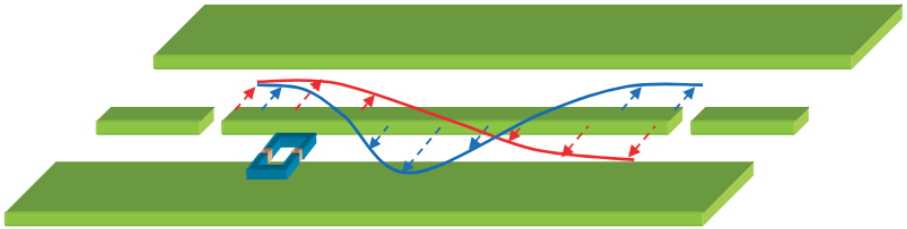


J. Aasi, *et al*, Nature Photonics, **7**, 613–619 (2013)
Dwyer, SE, *et al*, Galaxies 2022, **10**(2), 46, …

Existing methods



Parametric down-conversion
(nonlinear response is weak)

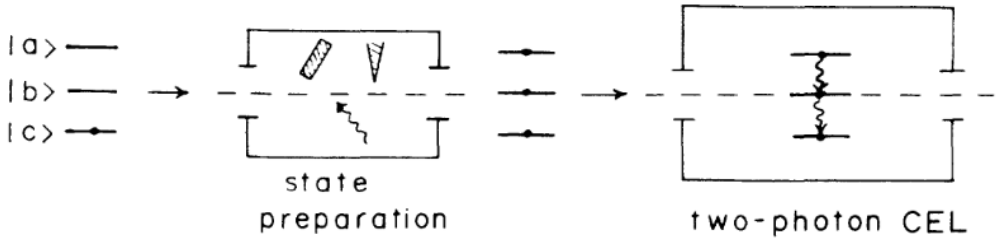


Ultra-strong coupling in cQED
(few microwave photons)

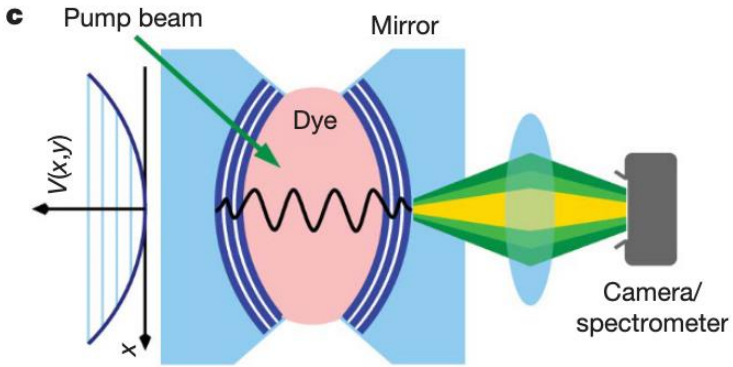
L.-A. Wu, *et al*, PRL **57**, 2520 (1986)
X. Guo, *et al*, Light: Science & Application **6**, e16249 (2017)
…

K. Moon, *et al*, PRL **95**, 140504 (2005)
A. Vaartjes, *et al*, Nature Commincations **15**, 4229 (2024)
Prof Jianqiang You's interesting talk
…

Existing methods



Two-photon correlated emission
(supress one-photon emission)

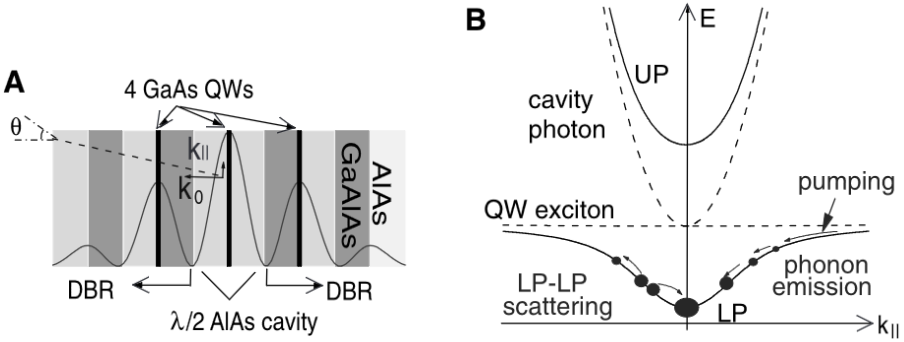


BEC in thermalized media
(not beyond optical coherence)

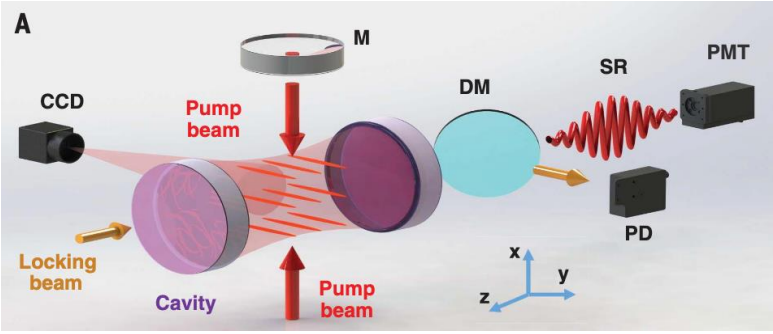
M. O. Scully, *et al*, PRL **55**, 2802 (1985)
D. J. Gauthier, *et al*, PRL **68**, 464 (1992)
…

J. Klaers, *et al*, Nature **468**, 545 (2010)
J. Schmitt, *et al*, PRL **112**, 030401 (2014)
…

New experimental insights



Exciton-polariton BEC



Pumped ultracold quantum gases
(Boson, Fermi, …)

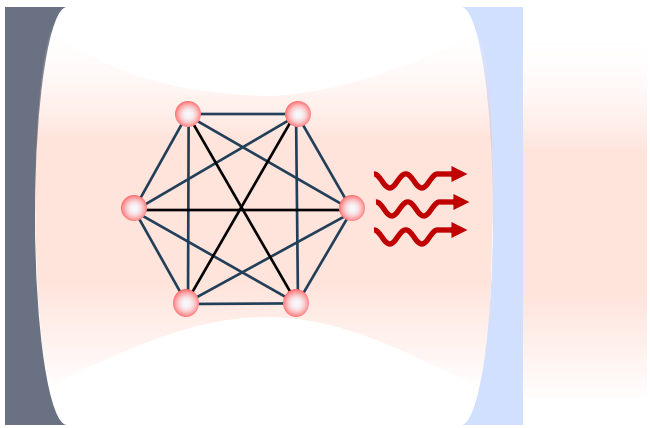
H. Deng, *et al*, Science **298**, 199 (2002)
A. Amo, *et al*, Nature **457**, 291 (2009)
Prof. Zhedong Zhang's interesting talk
…

K. Baumann, *et al*, Nature **464**, 1301 (2010)
X. Zhang, *et al*, Science **373**, 1359 (2021)
…

Correlations can transfer from 'matter' to light

Quantum many-body laser

KEY IDEA: Transfering the interaction-induced correlations from a quantum many-body emitter to photons through the lasing process.



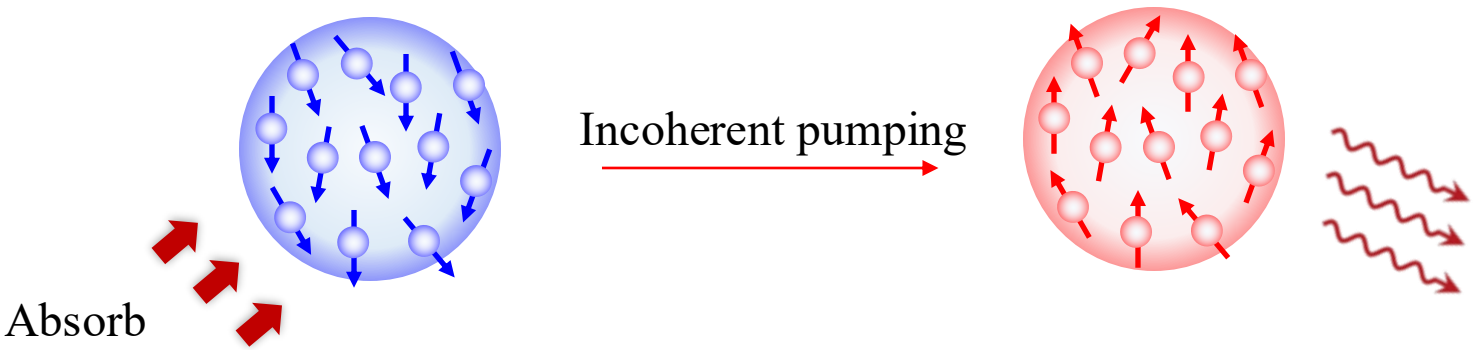
Optical coherence +

quantum correlations

*Recent advances in quantum technology have made coherently interacting emitters available (G. Bornet, *et al*, Nature **621**, 728–733 (2023), WJ Eckner, *et al*, Nature **621**, 734–739 (2023), …)

Necessary elements for a many-body laser

I. Pump and relaxation



- Incoherent pumping of emitters (rate w)

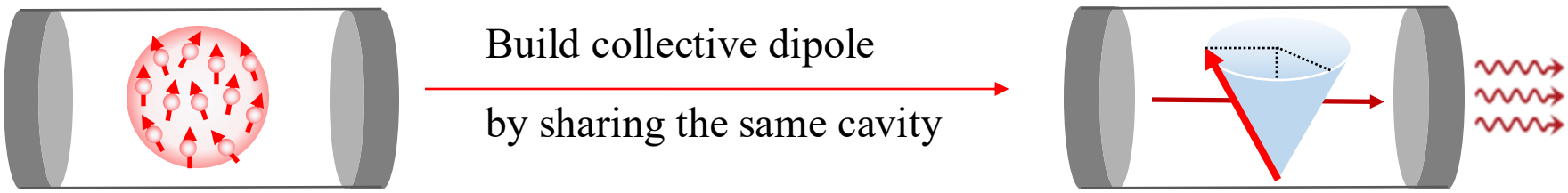
  — supply energy to emitters to achieve population inversion

- Relaxation of emitters and cavity (rate γ and κ)

  — transfer absorbed energy from emitters to light

Necessary elements for a many-body laser

II. Coherent emitter-light interaction



$$\hat{H}_{ac} = i\Omega\left(\hat{a}^\dagger \hat{J}^- - \hat{a}\hat{J}^+\right), \hat{J}^\pm = \sum_j \hat{\sigma}_j^\pm$$

Cooperativity: $$C = \frac{2N\Omega^2}{\kappa(w+\gamma)} \propto \frac{\sqrt{N}\Omega}{w+\gamma} \cdot \frac{\sqrt{N}\Omega}{\kappa}$$

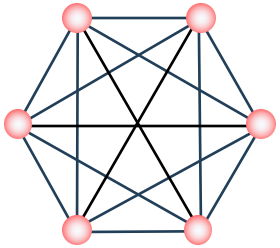
Collective dipole with locked phase — coherent emission

Necessary elements for a many-body laser

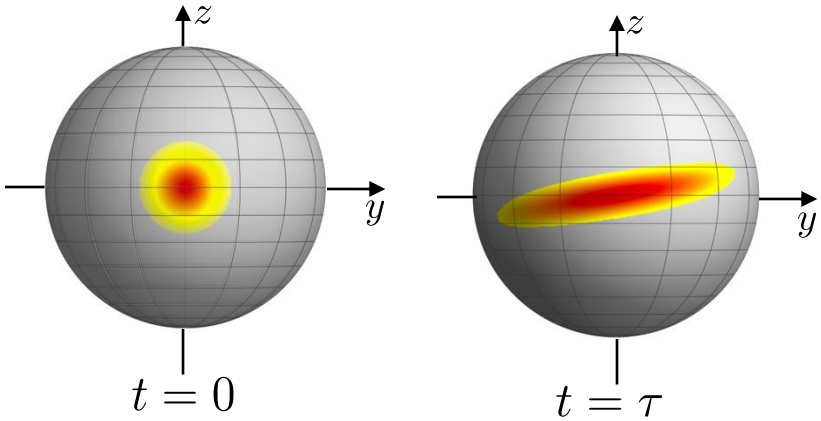
III. Quantum correlations within the quantum many-body emitter

$$\hat{H}_{aa} = \frac{\Delta}{2}\sum_{j=1}^{N}\hat{\sigma}_j^z - \epsilon\sum_{i<j}\hat{\sigma}_i^z\hat{\sigma}_j^z = \Delta\hat{J}^z - 2\epsilon\left(\hat{J}^z\right)^2$$

Lipkin-Meshkov-Glick (LMG) interaction



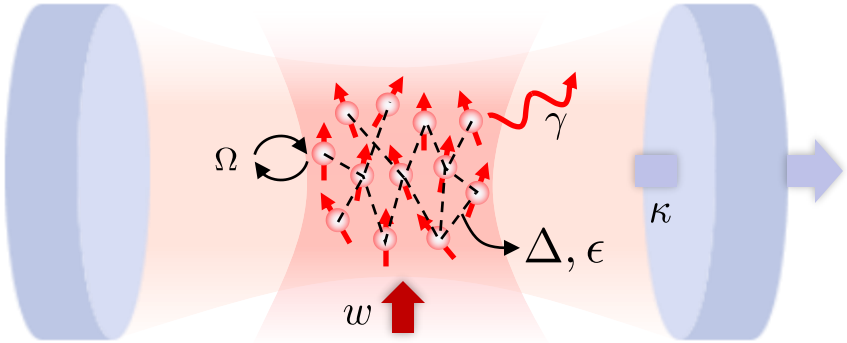
- Preserve the U(1) symmetry
- Preserve the permutation symmetry
- Produce squeezing correlations



M. Kitagawa, M. Ueda, Phys. Rev. A 47, 5138 (1993), …

Model

- ✓ Pumping
- ✓ Collectivity
- ✓ Interaction



$$\dot{\hat{\rho}} = \underbrace{-i\left[\hat{H}_{ac} + \hat{H}_{aa}, \hat{\rho}\right]}_{\text{coherent}} + \underbrace{w\sum_j \mathcal{L}_{\hat{\sigma}_j^+}[\hat{\rho}] + \gamma\sum_j \mathcal{L}_{\hat{\sigma}_j^-}[\hat{\rho}] + \kappa\mathcal{L}_{\hat{a}}[\hat{\rho}]}_{\text{incoherent}}$$

- ➢ $\hat{H}_{ac}$: building collectivity to lock phases of emitters (preserve correlations)
- ➢ Incoherent part: ensure the device have steady lasing
- ➢ $\hat{H}_{aa}$: producing quantum correlations within emitters

Small system: exact simulation

Permutation symmetry of Liouvillian:

$$\sigma\mathcal{L}\sigma^{-1} = \mathcal{L}, \qquad \sigma\left(|\uparrow,\downarrow,\uparrow,\cdots,\downarrow\rangle\right) \longrightarrow |\downarrow,\uparrow,\uparrow,\cdots,\downarrow\rangle$$

Dicke basis for atoms: $|J,M\rangle \equiv \frac{1}{\sqrt{D_J}}\sum_i |J,M,i\rangle$

In Dicke basis: $$\hat{\rho} = \sum_{n,n'}\sum_{J,M,M'} \rho^{n,n'}_{J,M,M'} |J,M\rangle\langle J,M'| \otimes |n\rangle\langle n'|$$

$$\mathcal{D}_{\hat{\rho}} = \frac{(N+1)(N+2)(N+3)}{6} N_{\text{cut}}^2$$

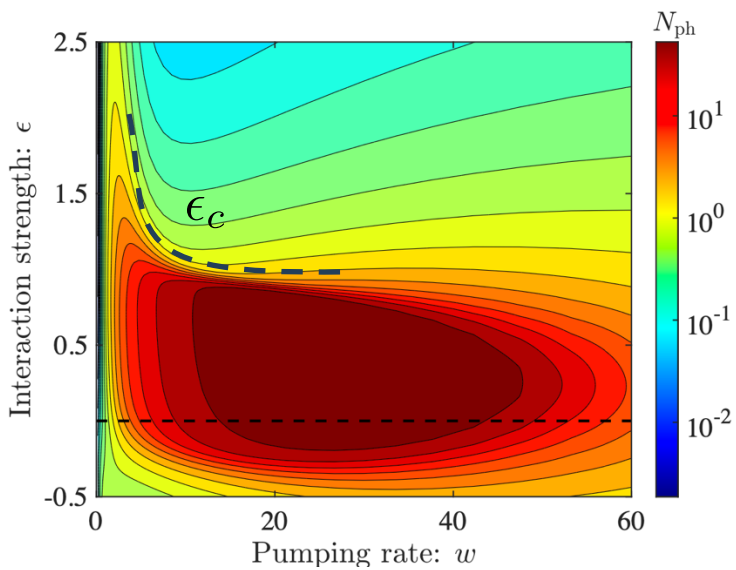
- $N_{\text{cut}} \sim o(N^2)$ for superradiance
- U(1) symmetry $M + n = M' + n'$

Total dimension of $\hat{\rho}$

$o(N^6)$

Phy. Rev. A, 78, 052101(2008),
New J. Phys. 20 ,112001 (2018),…

Small system: exact simulation



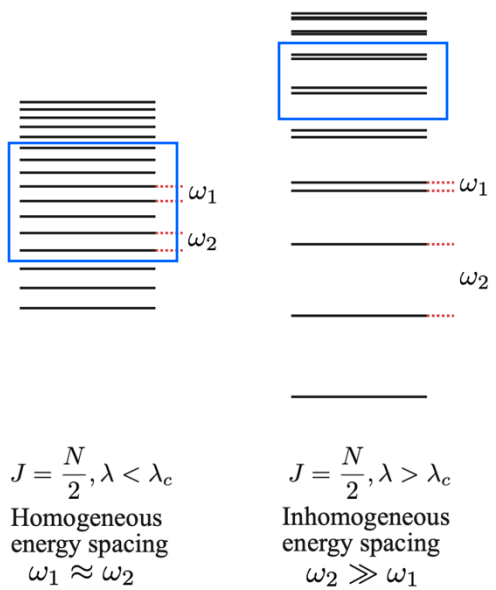
Average cavity photon for
$N = 15, N_{\rm cut} = 200$
$\kappa = 1, \gamma = 1, \Omega = 1$
$\Delta/N = 0.4$

Direct observations:

1. Photon burst at dark red region, indicating a lasing
2. Normal lasing threshold

   — population inversion: $w_{\rm min}$, fully polarized: $w_{\rm max}$
3. Lasing is blocked by strong interaction

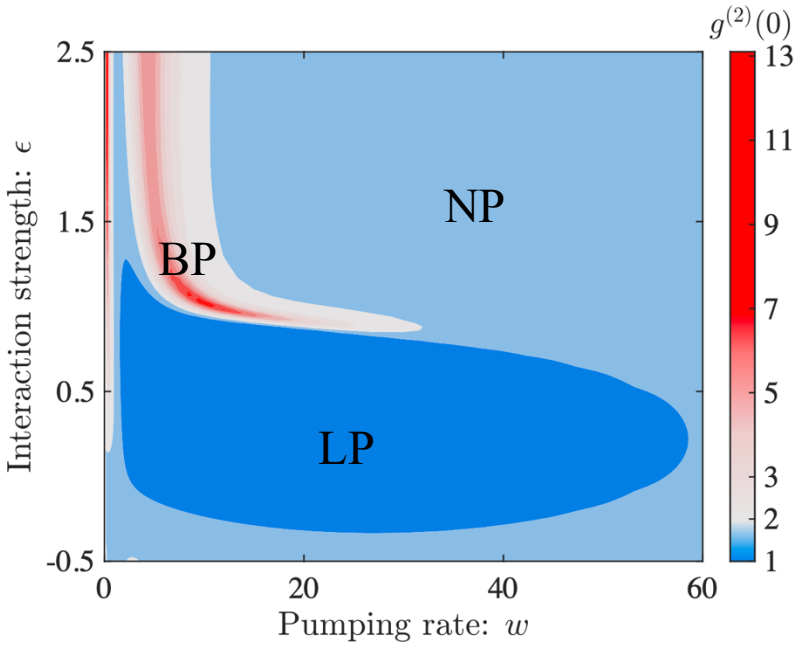
Lasing blockade mehacnism

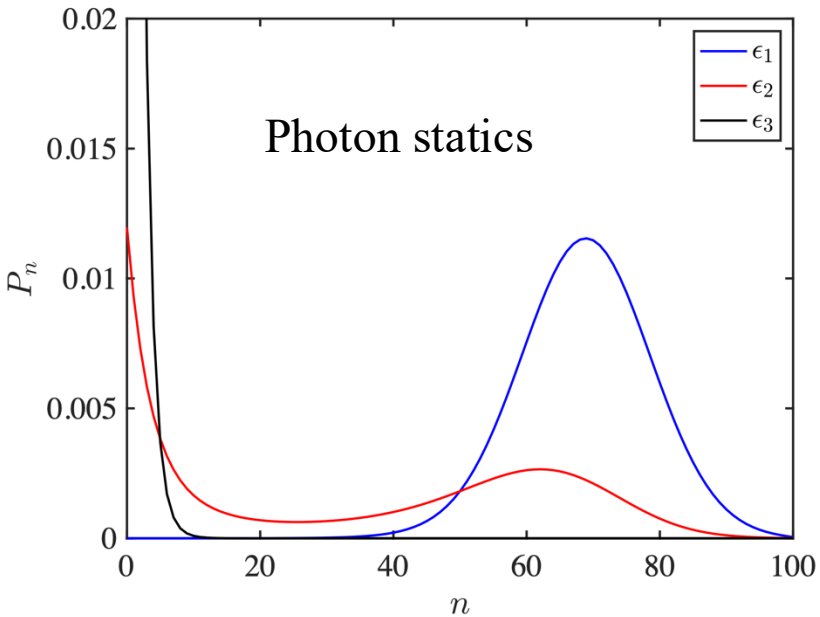


Light emission ⟷ Emitter's transition

- In the weak-interacting case, all emitters transition are near resonance with the cavity ⟶ **normal lasing**
- In the strong-interacting case, small part of emitters transition are near resonance with the cavity ⟶ **blockade lasing**
- Interaction also induce inhomogeneous energy spacing ⟶ **squeezing**

Interaction induced bistable phase





Phase diagram obtained from $g^{(2)}(0)$ and photon statics (interactions induced)

- Normal phase (NP: $g^{(2)}(0) = 2$)
- Lasing phase (LP: $g^{(2)}(0) = 1$)
- Bistable phase (BP: $g^{(2)}(0) > 1$)

Exact simulation limitations

- Numerical simulation complexity $\tilde{\mathcal{D}} \sim N^6$
- Finite size effect of the phase boundary
- Hidden squeezing correlation due to U(1) symmetry

Beyond exact simulation? → Mean field and fluctuation analysis

Mean field

$$\begin{cases} \dfrac{d\langle\hat{a}\rangle}{dt} = -\dfrac{1}{2}\left(\kappa + 2i\omega_c - 2i\omega\right)\langle\hat{a}\rangle + \ldots \\ \dfrac{d\left\langle\hat{J}^-\right\rangle}{dt} = 4i\epsilon\left\langle\hat{J}^z\hat{J}^-\right\rangle + \ldots \\ \dfrac{d\left\langle\hat{J}^z\right\rangle}{dt} = \ldots \end{cases}$$

*Negelect high-order fluctuations

Lasing condition ($\langle\hat{a}\rangle_s \neq 0$):

$$\underbrace{2C\frac{w-\gamma}{w+\gamma}}_{\mathscr{P}} > \underbrace{1 + \left(\frac{2\left(\Delta - 2\epsilon - 4\epsilon\left\langle\hat{J}^z\right\rangle_s\right)}{w+\gamma+\kappa}\right)^2}_{\mathscr{P}_c}$$

Lasing condition
- Population inversion
- Large cooperativity

$\xrightarrow{\mathscr{P}}$ Cooperativity*inversion factor

- Moderate interaction

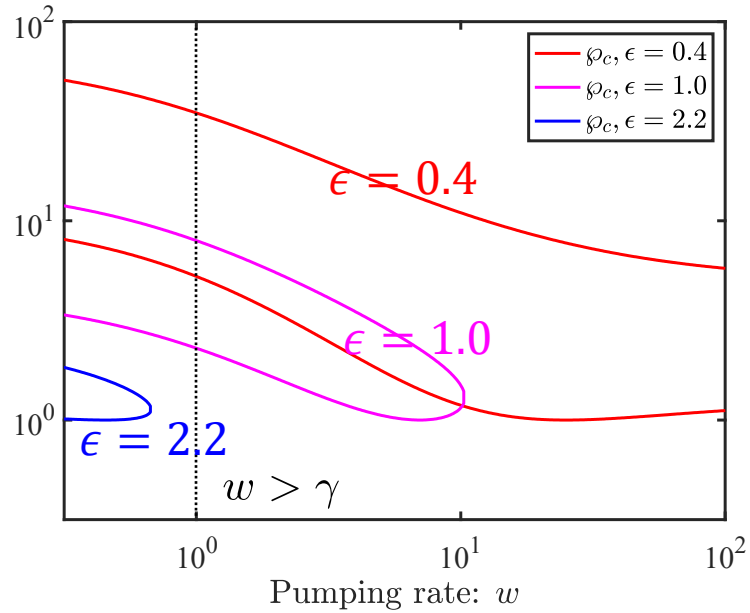
$\xrightarrow{\mathscr{P}_c}$ Lasing threshold

Lasing threshold

Lasing threshold for interacting system

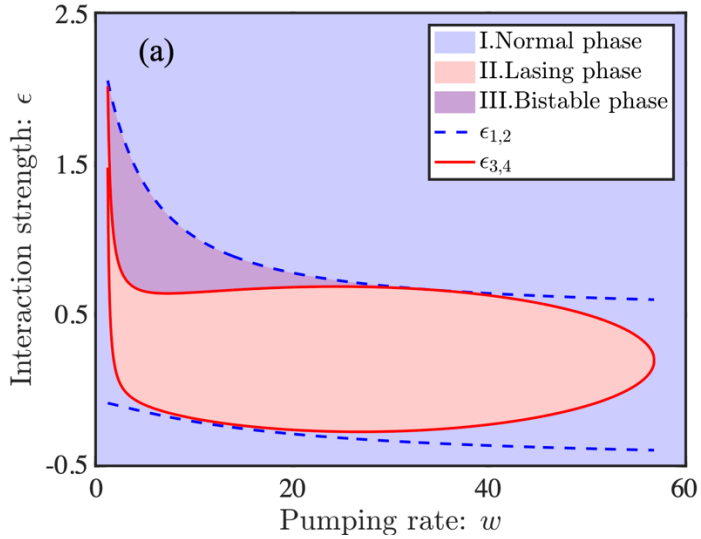
$$\mathscr{P}_c = 1 + \left(\frac{2\left(\Delta - 2\epsilon - 4\epsilon \left\langle \hat{J}^z \right\rangle_{\mathrm{s}}\right)}{w + \gamma + \kappa} \right)^2$$

- Self-consistent equation $\langle \hat{J}^z \rangle_{\mathrm{s}} = f(\mathscr{P}_c)$
- Nonlinearity increase as $|\epsilon|$

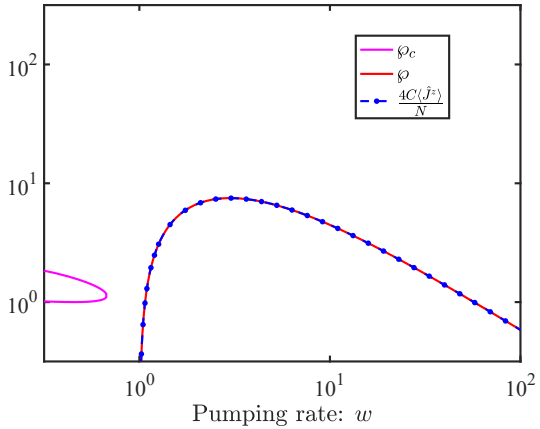


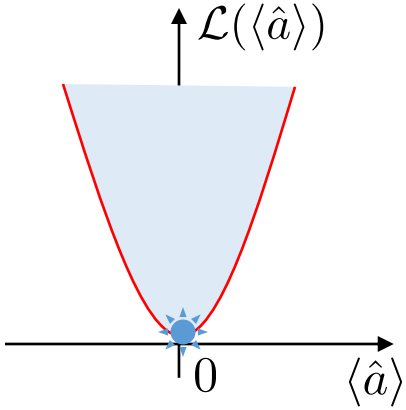
- Weak interacting → negligible nonlinearity ($\epsilon = 0$)
- Moderate interacting → strong nonlinearity → Bistable phase
- Strong interacting → blockade lasing

Phase diagram



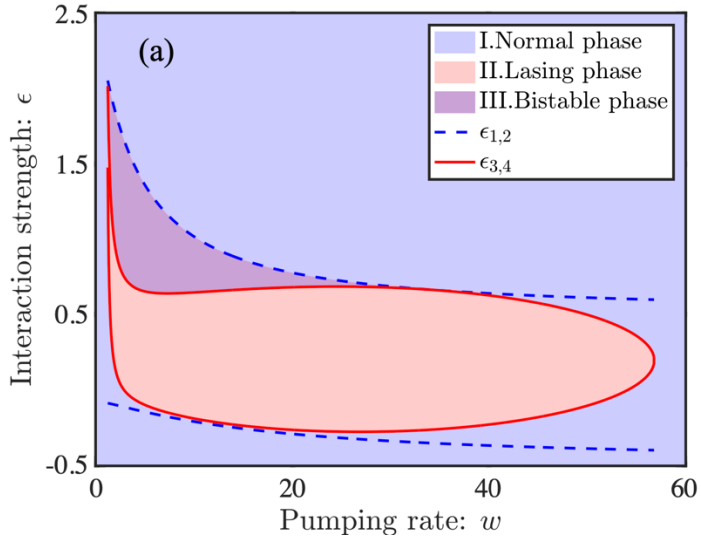
- Normal phase: no lasing



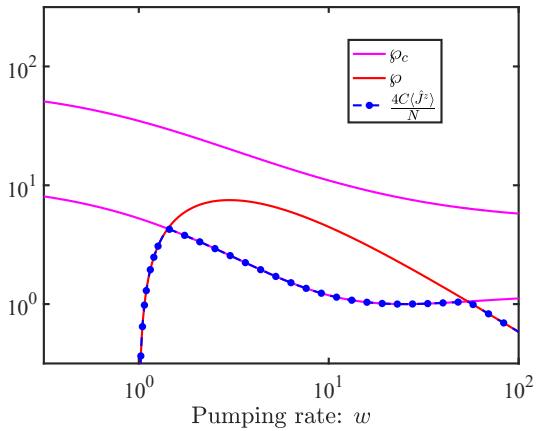


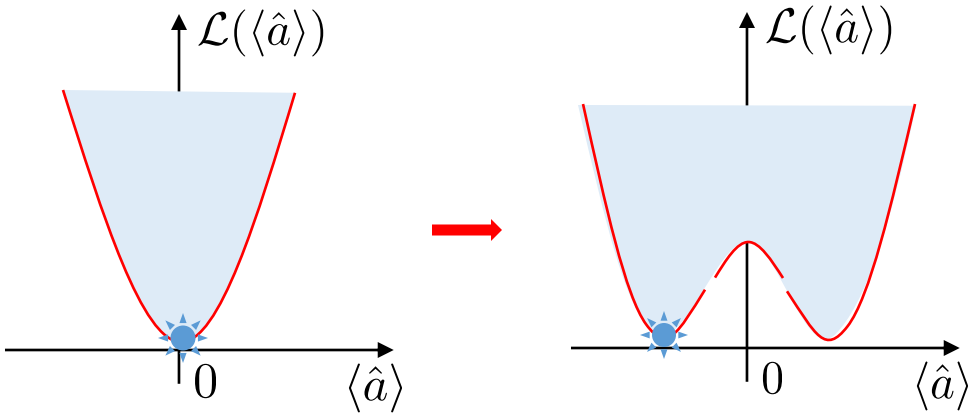
- No phase transition
- No U(1) symmetry breaking

Phase diagram



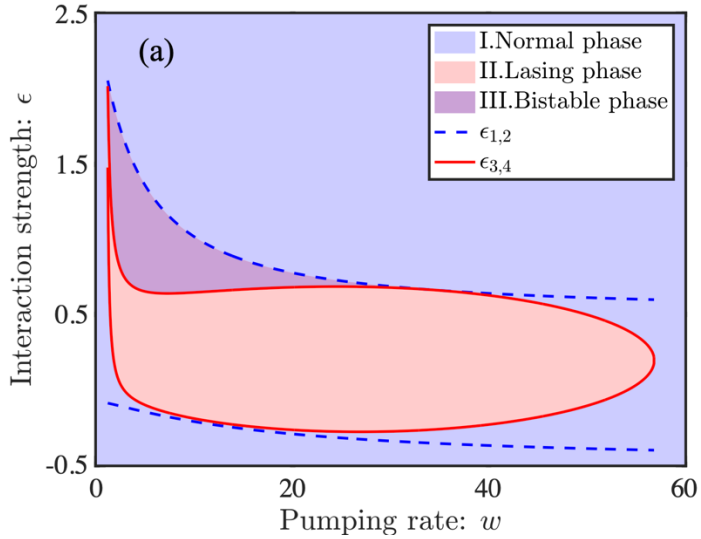
- Normal phase: no lasing
- Lasing phase: "traditional" lasing



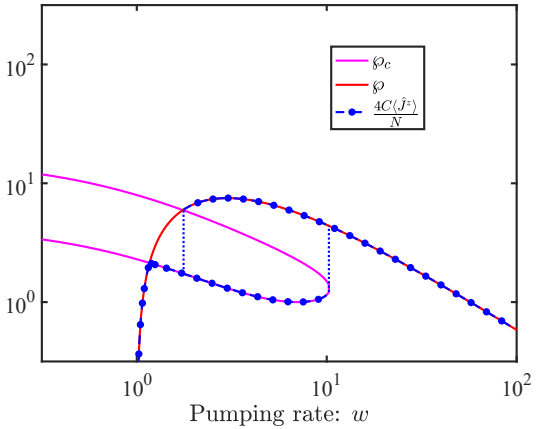


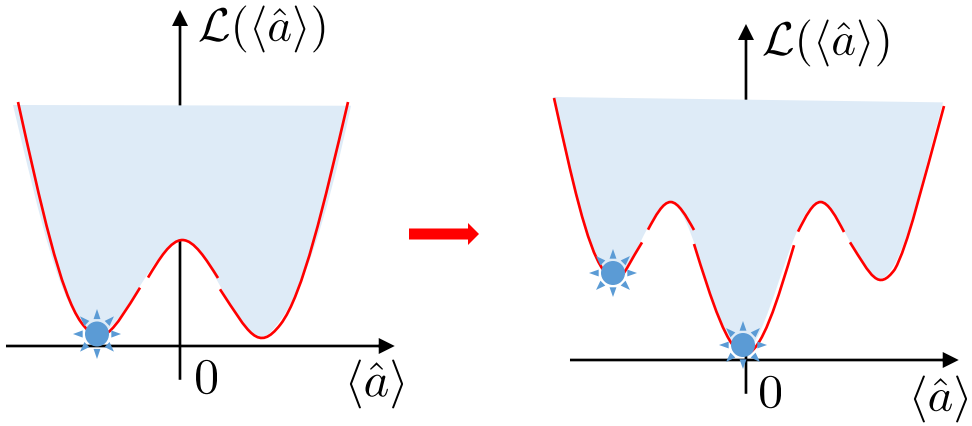
Phase transition and U(1) symmetry breaking

Phase diagram



- Normal phase: no lasing
- Lasing phase: "traditional" lasing
- Bistable phase: new lasing



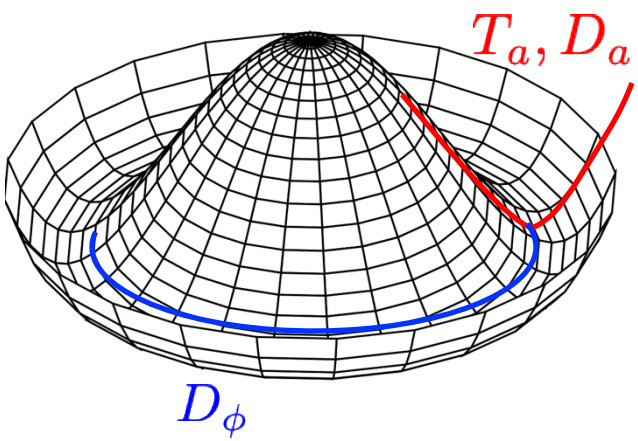


Bistability and first order phase transition

Fluctuation analysis

Effective Fokker-Planck equation for P-representation of photons:

$$\frac{\partial \tilde{P}}{\partial t} = \left(\frac{1}{T_a}\frac{\partial}{\partial z} z + \frac{1}{2D_a}\frac{\partial^2}{\partial z \partial z} + \frac{1}{2D_\phi}\frac{\partial^2}{\partial \phi \partial \phi} + \boxed{\frac{2\chi}{\mathcal{A}_s T_a}\frac{\partial}{\partial \phi} z - \frac{\chi}{2\mathcal{A}_s T_a}\frac{\partial^2}{\partial z \partial \phi}} \right) \tilde{P}$$



When lasing $\mathscr{P} > \mathscr{P}_c$

Fast amplitude damping, long phase diffusing

$$D_\phi \gg T_a$$

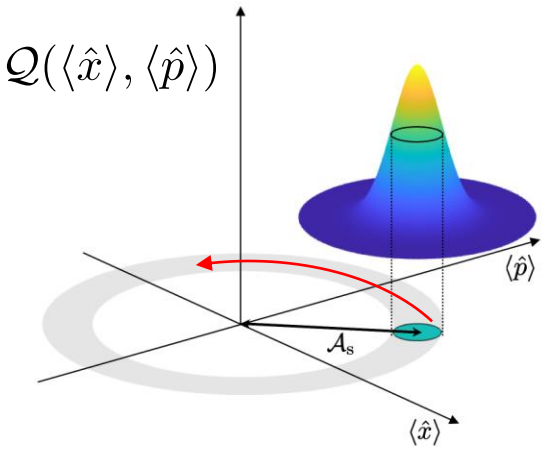
Additional correlations between amplitdue and phase due to interactions among atoms

$$\chi = \frac{2N\epsilon\mathscr{P}_c}{C(w+\gamma+\kappa)}$$ Interaction effects

Carmichael, Howard J. Statistical methods in quantum optics. Springer Science & Business Media, 2013.

Number-phase squeezing

Without interaction

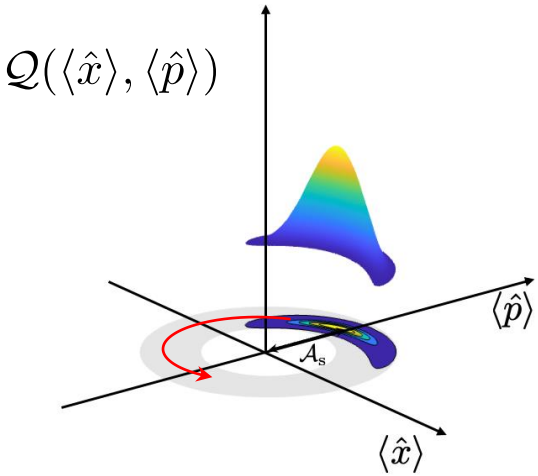


The phase fluctuation is irrelevant to the amplitude flucutaitons

$$\phi \not\leftrightarrow z$$

Normal Lasing: **coherent state**

With interaction

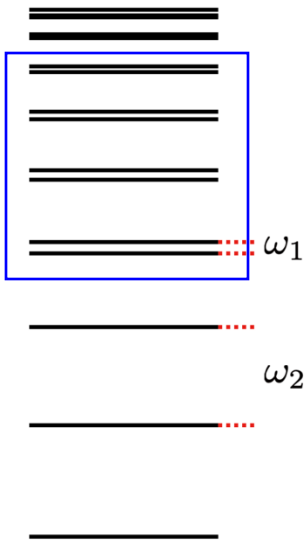


The phase fluctuation is correlated to the amplitude flucutaitons:

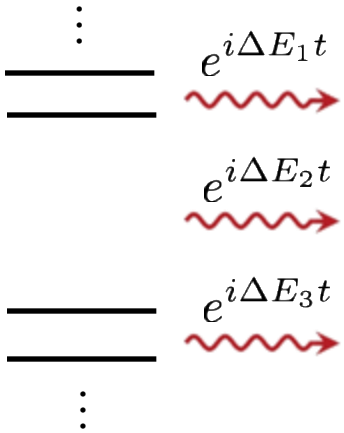
$$\phi \xleftrightarrow{\chi} z$$

New Lasing: **squeezed state**

Squeezing mechanism

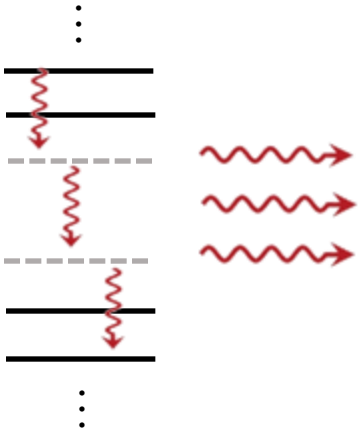


Strong interacting case



Phase viewpoint

Nonlinear-phase modulation

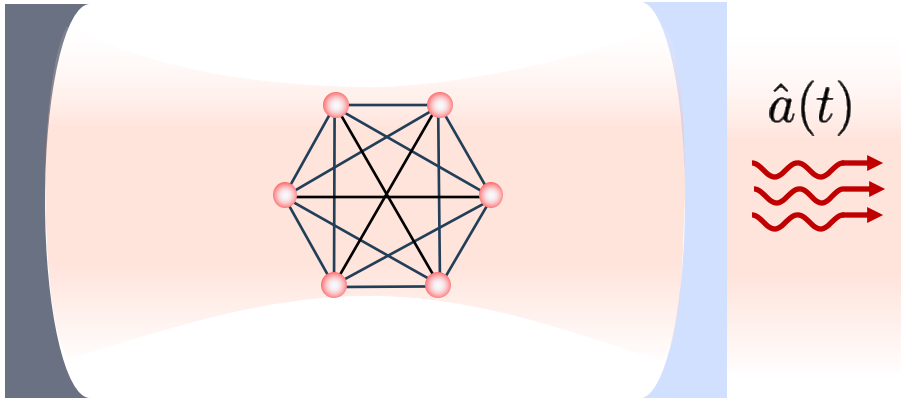


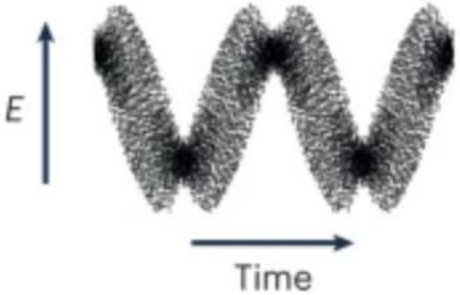
Photon-number viewpoint

Multi-photon bundled emission

Number-phase squeezing

Quantify squeezing





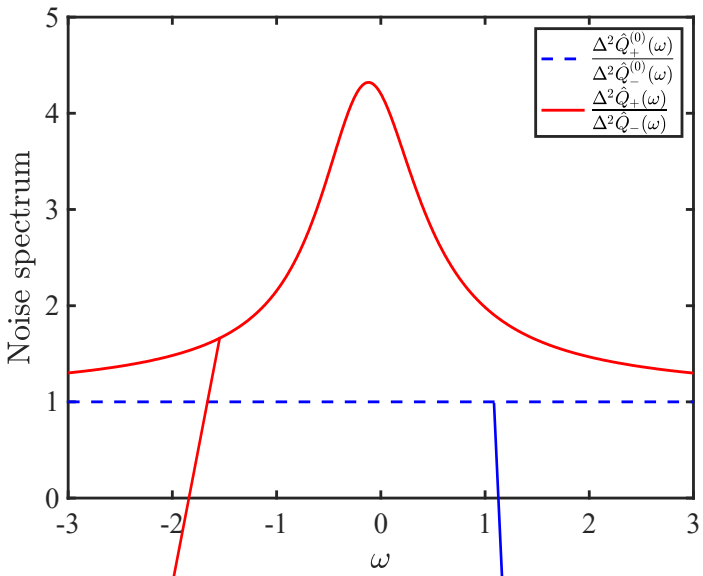
example

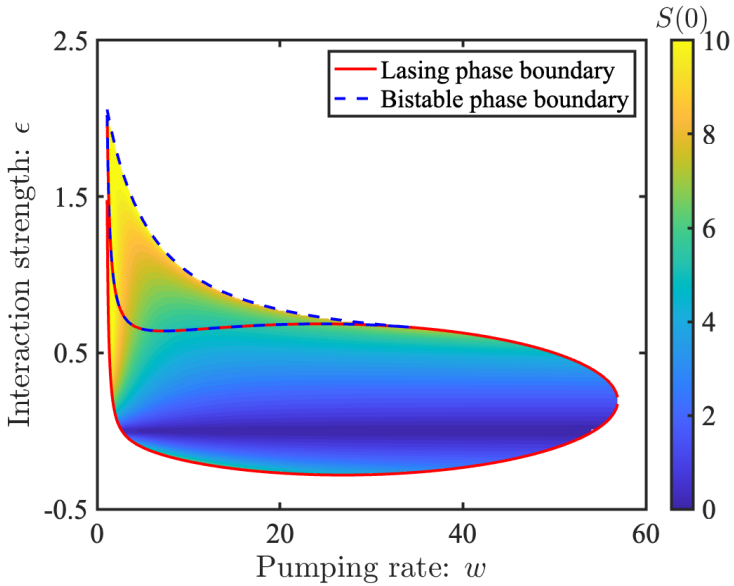
Input-output theory $\longrightarrow$ quadurature spectrum $\Delta^2\hat{Q}_+(\omega), \Delta^2\hat{Q}_-(\omega)$

Squeezing $S(\omega) = \dfrac{\Delta^2\hat{Q}_+(\omega)}{\Delta^2\hat{Q}_-(\omega)}$ $\longrightarrow$ Static squeezing $S(0)$

$$\boxed{S(0) = 40\,\mathrm{arcsinh}(\chi)/\ln(10), \chi \sim \frac{\kappa \mathscr{P}_c \epsilon}{2\Omega^2}}$$

Noise spectrum and squeezing paramter





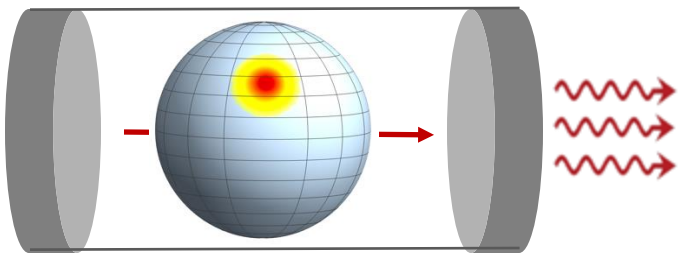
Coherent state
Noninteracting ($\epsilon = 0$)

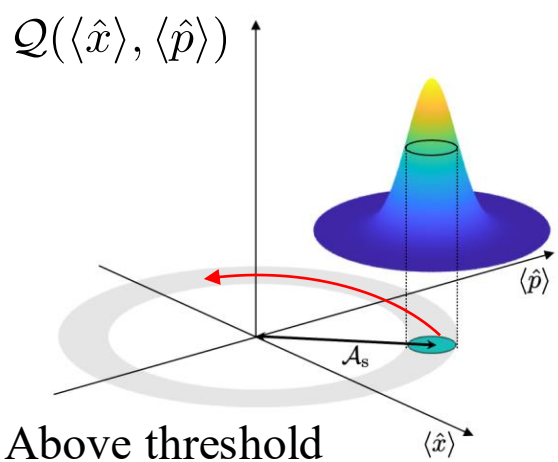
Squeezed state
Interacing ($\epsilon \neq 0$)

- About 5dB within the bistable phase (choosen parameter)
- Scaling $S(0) \sim 40\ln(\epsilon)/\ln(10)$

Compared with traditional superradiant lasing

Traditional superradiant lasing

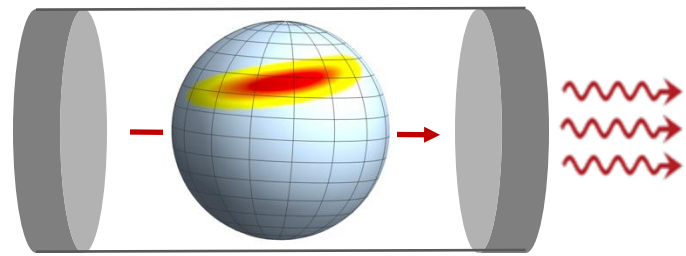


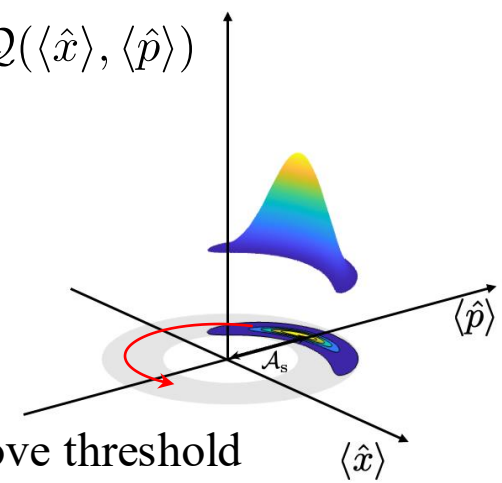


Above threshold

$\mathscr{P} > \mathscr{P}_c$

Superradiant squeezed lasing



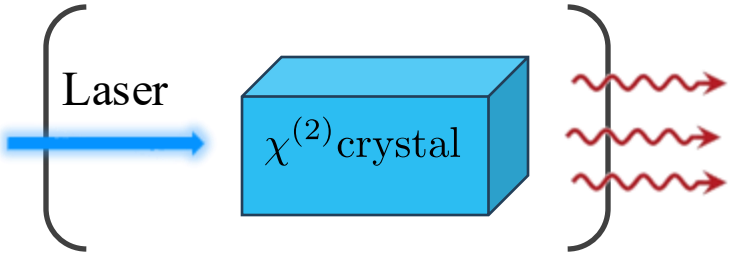


Above threshold

$\mathscr{P} > \mathscr{P}_c$

Compared with PDC squeezing

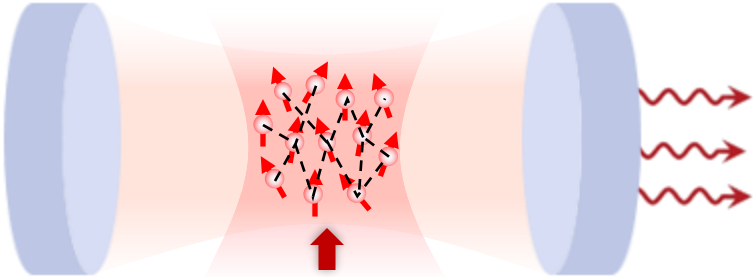
PDC squeezing



Coherent inputs

$$\hat{H}_{\text{eff}} = \chi^{(2)} \left((\hat{a}^\dagger)^2 \hat{b} + \hat{a}^2 \hat{b}^\dagger \right)$$

Superradiant squeezing



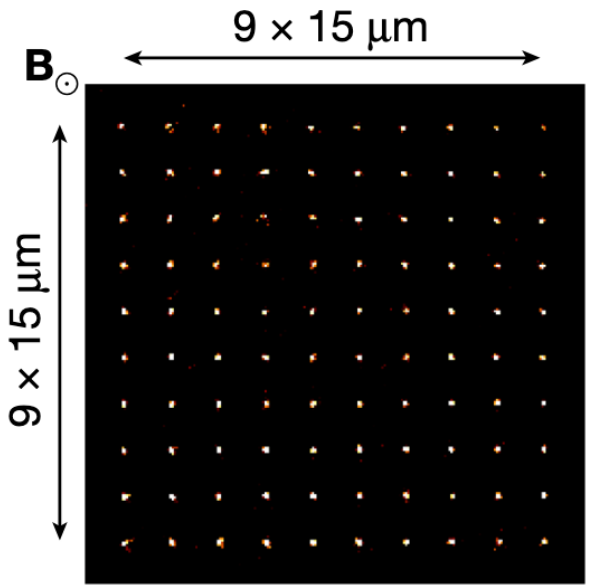
Incoherent pumping

$$\hat{H}_{\text{eff}} = \chi_{\text{eff}} \left(\hat{a}^\dagger \hat{a} - \mathcal{A}_{\text{s}}^2 \right)^2$$

A new and controllable mechanism to generate quantum light

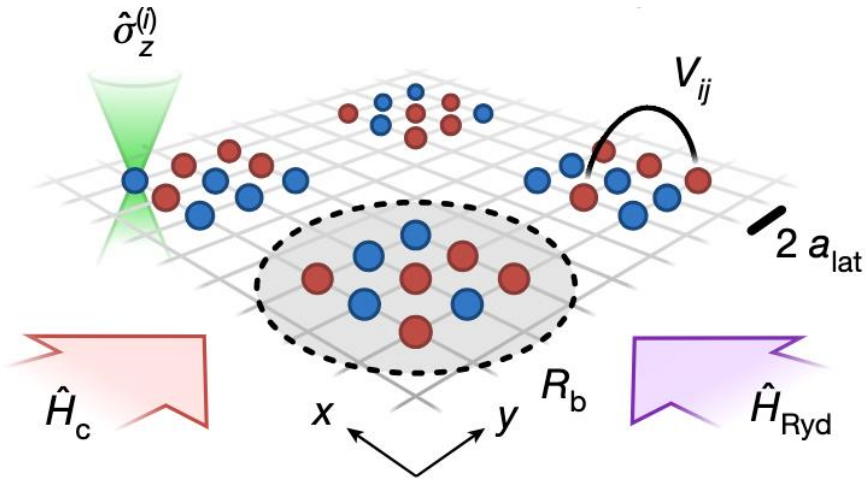
M. Kitagama, *et al*, Phys. Rev. A, **34**, 3974(1986)

Experiment proposals and predications



Dipolar Rydberg atom array

Nature **621**, 728–733 (2023)



Ultracold atoms array

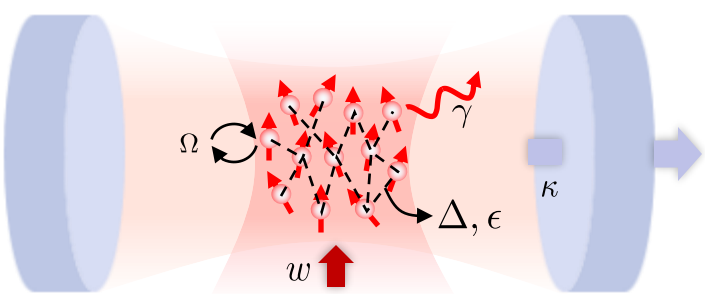
Nature **621**, 734–739 (2023)

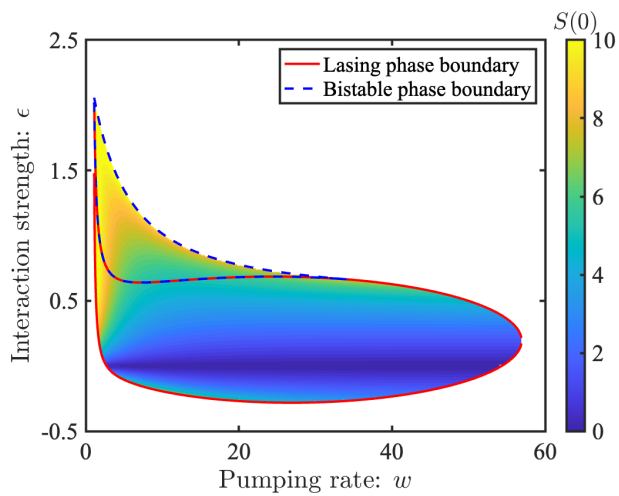
Experiment proposals and predications

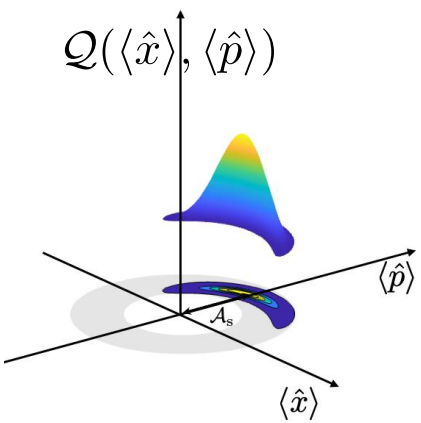
- Photon fluctuations $g^{(2)}(0) = \langle (\hat{a}^\dagger)^2 \hat{a}^2 \rangle_{\mathrm{s}} / \langle \hat{a}^\dagger \hat{a} \rangle_{\mathrm{s}}^2$
- Phase diagram and bistable phase $\mathscr{P}_c = 1 + \left(\frac{2\left(\Delta - 2\epsilon - 4\epsilon \left\langle \hat{J}^z \right\rangle_{\mathrm{s}}\right)}{w + \gamma + \kappa} \right)^2$
- Steady number-phase squeezed light $S(0) = 40 \operatorname{arcsinh}(\chi)/\ln(10)$
- …

Summary

- Lasing can be blockade due to strong interactions within emitters
- A new bistable phase emerges under moderate interactions, accompanied by a first order phase transition.
- In this new bistable phase, light emission exhibits strong number-phase squeezing, giving a new mechnism to produce **quantum light**.







Conclusions

Traditional laser
(>1960s)

One atom
(independent atoms)

Towns,
Haken,
…

Superradiant laser
(>1970s)

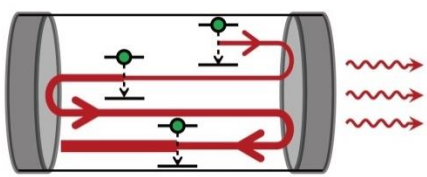
N-noninteracting
atoms

Dicke,
M. J. Holland,
…

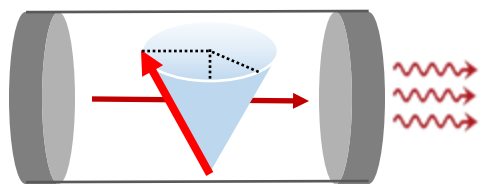
Many-body laser
(ongoing)

N-coherent
interacting atoms

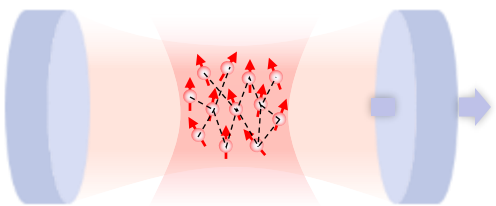
*This work



Frequency pin to cavity
$T_\phi = N_c \tau_c$



Frequency pin to atoms
$T_\phi = N_a \tau_a$



Coherence + quantumness

Da-Wu Xiao, Chong Chen, & Ren-Bao Liu (In preparation)